Adaptive Cognitive-Load Study Timer for Personalized Break Scheduling

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Proposal Document

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Declaration

I declare that this is my own work and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Abstract

Effective study planning is often hindered by the lack of awareness of individual productivity rhythms, leading students to study at suboptimal times and experience reduced focus. Traditional study tools provide basic timers and reminders but rarely adapt to personal chronotypes or contextual factors such as sleep and session length. To address these limitations, this project introduces a novel study support system that combines chronotype heatmap visualization with a contextual bandit–based recommendation model to give insightful advice to users to optimize their study plans and be more productive.

The proposed system monitors in-app activity, including keystrokes, scrolls, session timings, and micro-EMA prompts (e.g., self-reported focus levels according to session times and session longevity), to generate personalized heatmaps of productivity across different times of the day and week. Building on these insights, a hybrid contextual bandit model adapts recommendations by integrating learned chronotype priors with real-time behavioral data, thereby providing more accurate and personalized scheduling advice to users.

The methodology involves developing the heatmap-based tracking interface, recommendation system, implementing the hybrid bandit model, and testing the system with students to evaluate usability, adaptability, and learning outcomes. Expected results include enhanced focus by identifying optimal study windows, improved time management through adaptive recommendations, and contextual insights such as the impact of sleep duration or session length on productivity etc.

By moving beyond simple visualization toward adaptive behavioral analysis, this study tool provides a personalized and scientifically grounded approach to improving student learning efficiency.

Key Words : cognitive load, micro-EMA, chronotypes, contextual bandit

Table of Contents

Supplementary Sections

List of Figures

List of Tables

1. Introduction

Students frequently encounter difficulties in determining optimal study periods, which can diminish concentration and reduce learning efficiency. Conventional study-support applications typically rely on static timers or reminders; however, these approaches lack adaptability to individual productivity rhythms, often referred to as chronotypes. Consequently, learners may engage in study sessions during non-ideal hours, such as late at night, resulting in diminished focus and information retention.

This study addresses these challenges within the domain of *Educational Technology and Machine Learning*, applying advanced computational methods to personalized learning. The proposed framework integrates chronotype-based heatmap visualization with *a contextual bandit model* for adaptive scheduling. The system collects and analyzes behavioral data—including keystroke activity, scrolling patterns, session durations, and short self-reported focus levels—to construct personalized study profiles. These technologies were chosen for their capacity to dynamically model individual productivity patterns and provide actionable, real-time recommendations. Unlike conventional tools, this approach aligns study intervals with students' natural rhythms, enhancing both focus and retention.

The objective of this research is to develop a system that not only monitors study behavior but also generates data-driven, personalized recommendations to improve concentration, learning efficiency, and overall academic performance. By explicitly combining machine learning techniques, adaptive algorithms, and behavioral analytics, this work demonstrates a novel application of key technologies in the specialized area of educational personalization.

1.1 Background and Literature Survey

Conventional productivity and study-support applications often adopt a uniform model of time management, recording only elementary indicators such as total study duration or task completion time. Such designs fail to account for daily fluctuations in cognitive capacity, which are strongly influenced by circadian rhythms and individual chronotypes. Neglecting these factors can result in poorly aligned study schedules and diminished learning outcomes, ultimately reducing overall academic achievement.

Chronotype, defined as an individual's inherent tendency toward morningness or eveningness, has been recognized as an influential determinant of cognitive performance and academic outcomes. Zerbini *et al.* [1] provided early evidence of its educational significance, demonstrating clear associations between chronotype and student achievement. Expanding on this theoretical basis, Goldstein *et al.* [2] empirically confirmed the "synchrony effect," indicating that students achieve higher scores on cognitive assessments when evaluated during their preferred time of day. Similarly, a meta-analysis by Preckel *et al.* [3] substantiated these observations, highlighting consistent correlations between chronotype alignment and academic success across diverse learning contexts.

Recent longitudinal investigations have provided further support for chronotype-sensitive educational practices. Vollmer *et al.* [4] reported that improved alignment between student chronotypes and institutional schedules was associated with reduced grade retention, underscoring the practical value of personalized timing strategies. In parallel, Goldin *et al.* [5] examined the mechanisms underlying chronotype misalignment and its detrimental effects on academic outcomes, emphasizing the importance of adaptive technologies in mitigating these challenges.

To address the challenge of personalizing learning schedules, contextual bandit algorithms have emerged as a promising approach for adaptive recommendation systems. Chen et al. (2023) developed contextual multi-armed bandit frameworks specifically for adaptive sequence learning, providing the algorithmic foundation for real-time educational personalization [6]. Lan et al. (2016) pioneered the application of contextual bandits in educational settings, demonstrating their effectiveness in personalized learning action selection [7]. More recently, Hunziker et al. (2020) implemented reinforcement learning approaches for adaptive scheduling of educational activities, showing significant improvements in learning outcomes through personalized timing recommendations [8].

The integration of contextual bandits with educational technology has shown particular promise in large-scale applications. Recent surveys on multi-armed bandit algorithms for adaptive learning highlight their effectiveness in providing personalized feedback and recommendations to students [9]. Advanced implementations have successfully deployed contextual bandits in large-scale online tutoring systems, optimizing feedback for over one million students and demonstrating the scalability of such approaches [10].

Visualizing temporal performance patterns has been identified as a critical factor in promoting user comprehension and system adoption. Prior research in productivity visualization has demonstrated that heatmaps are effective for representing time-dependent performance trends, employing color gradients to emphasize periods of peak and low activity [11]. Practical applications in workforce management, such as Time Champ’s lifetime heatmaps for employee performance analysis, have highlighted the utility of temporal visualization for decision-making [12]. Comparable strategies in habit-tracking applications further illustrate the ability of heatmaps to convey behavioral patterns clearly and intuitively to users [13].

The convergence of chronotype research, contextual bandit algorithms, and visualization techniques presents an opportunity for developing sophisticated adaptive learning systems. However, existing research has primarily focused on these areas independently, with limited integration of chronotype awareness into adaptive scheduling algorithms. Most productivity applications continue to ignore individual circadian preferences, while educational research on chronotypes has not been translated into practical technological solutions.

To address this limitation, the present study proposes a chronotype-based heatmap system that integrates empirical findings on chronotypes with contextual bandit algorithms for adaptive scheduling. The system is designed to deliver personalized insights into study performance—for example, indicating that shorter, 50-minute sessions may yield up to 30% higher efficiency compared to longer, 120-minute sessions—by analyzing individual behavioral patterns in real time. By combining predictive modeling with intuitive heatmap visualization, the platform facilitates data-driven decision-making regarding optimal study periods, thereby addressing a key challenge in personalized educational technology.

1.2 Research Gap

While chronotype research has been extensively studied in educational psychology and contextual bandit algorithms have proven effective in various recommendation systems [6] [7] [8], existing study timer applications fail to integrate these insights into practical productivity tools. Current popular study applications such as Forest, Be Focused, Toggl Track, and traditional Pomodoro timer apps predominantly focus on basic time tracking, task categorization, and static session management without incorporating individual chronotype patterns or adaptive scheduling capabilities [9] [10]. These applications operate on fixed timer intervals (typically 25-minute Pomodoro sessions) and assume uniform productivity across all users and time periods, ignoring the substantial body of research demonstrating significant individual variations in cognitive performance throughout the day.

Moreover, existing productivity applications lack sophisticated features such as chronotype heatmaps that visualize individual performance patterns, adaptive timers that adjust session durations based on real-time cognitive load assessment, and personalized insight generators that provide quantified recommendations. While applications like RescueTime and Clockify offer retrospective analytics, they do not incorporate predictive modeling or contextual bandit algorithms to optimize future study sessions based on individual circadian rhythms and historical performance data. The absence of cognitive load calculators in current timer applications represents a significant limitation, as these tools continue to employ one-size-fits-all approaches that fail to account for the dynamic nature of human attention and energy levels throughout different times of day and week.

The gap lies in the lack of intelligent study timer applications that combine chronotype awareness with adaptive scheduling algorithms and real-time cognitive load assessment. Current study tools, including those used in productivity tracking and time management, do not fully address the complexity of individual circadian patterns and their impact on learning effectiveness. Additionally, while basic visualization techniques exist in current productivity apps, their application to chronotype-based performance prediction with actionable, personalized insights such as "You perform 30% better in 50-minute sessions than 120-minute sessions during morning hours" has not been implemented in existing commercial applications.

This study aims to fill this gap by developing an intelligent chronotype heatmap system integrated within a study timer application that specifically combines individual circadian rhythm analysis with contextual bandit algorithms and cognitive load calculators. Unlike existing timer applications that rely on static intervals and basic tracking, this system will provide adaptive session recommendations, predictive performance heatmaps, and personalized scheduling insights based on continuous learning from individual usage patterns. By addressing the specific limitations of current study timer applications and integrating advanced chronotype research with practical educational technology, this research will contribute to revolutionizing personalized productivity tools and advancing the effectiveness of self-regulated learning systems.

1.3 Research Problem

The central challenge addressed in this study is the absence of personalized study-support systems that adapt to individual chronotypes and attention rhythms. Current tools—including timers, planners, and static recommendation systems—typically employ a uniform approach, overlooking the natural variation in cognitive performance among students. As a result, learners often engage in study sessions at suboptimal times, leading to decreased efficiency, lower retention, and wasted effort.

Several commercial applications attempt to incorporate aspects of chronotype or productivity awareness. For example, the *Good Morning Sunshine app* provides general chronotype-based recommendations for daily productivity, *Oura Ring's Body Clock feature* offers insights into sleep patterns and individual circadian preferences, and *Doodle* allows chronotype-informed scheduling for meetings. While these products offer valuable information, they primarily focus on general wellness or scheduling and do not integrate detailed study behavior analytics or adaptive learning algorithms. Consequently, learners still lack dynamic, personalized study guidance.

Additionally, existing systems that employ contextual bandit algorithms for personalization face challenges such as the cold-start problem, where insufficient data in early use limits the accuracy of recommendations. Most solutions also fail to combine objective behavioral metrics—such as typing activity, scrolling patterns, or session duration—with subjective self-reports of focus. Without this integration, it is not possible to construct reliable chronotype-based profiles or adapt dynamically to changes in attention over time.

To address these gaps, this research proposes a hybrid system that combines focus mapping with a modified contextual bandit algorithm. By leveraging both behavioral and self-reported inputs, the system can continuously identify optimal study periods for each student and generate adaptive recommendations that improve with ongoing use. This approach not only mitigates the cold-start problem but also ensures that study schedules align with individual chronotypes and attention rhythms, thereby enhancing personalization, learning efficiency, and overall academic performance.

2. Objectives

2.1 Main Objectives

The primary objective of this research is to develop a smart study timer application that integrates behavioral data, measures cognitive load through user interactions, and incorporates self-reported focus levels across varying session lengths and times. A contextual bandit algorithm is employed to generate personalized study recommendations aligned with each student's chronotype and feedback. The system is designed to address the inefficiencies and unreliability of existing study timers, providing dynamic, data-driven guidance that enhances students' focus, engagement, and overall learning outcomes.

2.2 Specific Objectives

1. **To develop a hybrid focus-mapping system that integrates contextual bandit algorithms with chronotype-based personalization.**

- **model Selection:** Choose suitable algorithms (contextual bandits: LinUCB, Thompson Sampling) and chronotype modeling methods (questionnaire or sleep-tracking).
- **Integration:** Merge chronotype adjustments with contextual bandit logic into a hybrid model; design simulations to test system logic before deployment.

2. **To evaluate and tweak the system's performance:**

- **Testing with student and users:** Testing system with real users to see systems adaptability and insight accuracy
- **Tweak system:** Reconsider selected models and application logics accoding to results of usage

3. **Design a User-Friendly Interface:**

- **Interface Development:** Create user friendly heatmaps and visualization for get necessary decision for their studying schedules for be more productive
- **Usability Testing:** Testing implemented user interface with selected individuals for testing how it affect to identify productive sessions

4. **To assess the impact on learners:**

- **Select areas:** Select study impact measures (focus variance, study efficiency, engagement metrics)
- **Optimization:** Analyze test results foe optimize system

5. **Evaluate and Document System Performance:**

- **Performance Metrics:** Establish performance metrics to evaluate the effectiveness of the recommendation system, including reliability, accuracy, and user satisfaction.
- **Documentation:** Prepare documentation for the system, including technical specifications, user manuals to support deployment and future enhancements.

3. Methodology

This system is designed to develop a dynamically responsive study timer that adapts to students' individual chronotypes and study habits by monitoring cognitive load and behavioral patterns. Data collection is performed through sensors and behavioral tracking mechanisms, capturing metrics such as keystrokes, scrolling activity, session durations, and self-reported focus levels. The collected data is subsequently analyzed to visualize students' study behaviors, highlighting the periods, conditions, and durations during which they exhibit peak productivity.

The system utilizes these insights to deliver personalized, data-driven study recommendations that enhance learning efficiency and academic performance. Development of the system involves several key steps, including the training of a contextual bandit model, integration of historical chronotype data, and iterative testing and evaluation with student participants. Through this workflow, the system continuously adapts to individual behaviors and cognitive patterns, providing real-time, tailored guidance that aligns with each student's unique productivity rhythms.

Several machine learning approaches were evaluated for adaptive scheduling, including Supervised Learning, Reinforcement Learning, and Contextual Bandits. As illustrated in Figure 3.1, Contextual Bandits provide a favorable balance of high personalization and moderate computational cost, making them particularly suitable for real-time study timer recommendations.

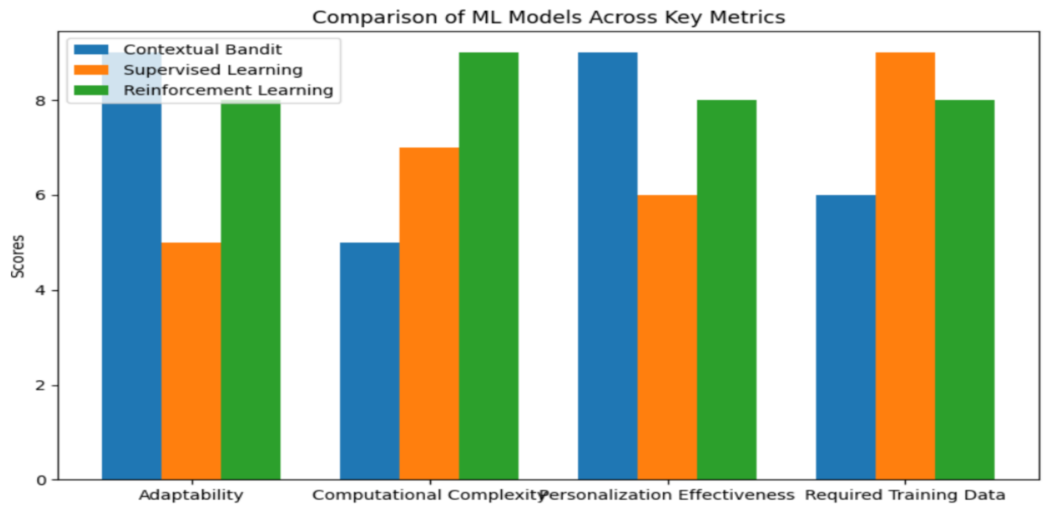

figure 3. 1 – Comparison of machine learning approaches for recommendation system

3.1 Overall System Description

1. **System Architecture:**

- **Chronotype Heatmap Module:** This module collects and processes user activity data (such as sleep-wake cycles, productivity logs, or app usage patterns) to generate personalized chronotype heatmaps. These heatmaps represent the user's natural rhythms and provide priors for the learning model.

- **Contextual Bandit Engine:** At the core of the system is a contextual bandit model that uses the chronotype heatmap as initialization. It continuously learns by interacting with users in real time, balancing exploration of new scheduling options with exploitation of known effective time slots.

- **Feedback Loop:** Users provide implicit (e.g., task completion rates, engagement levels) and explicit (e.g., satisfaction ratings) feedback. This feedback is processed by the contextual bandit engine to refine schedules dynamically.

- **User Interface (Mobile/Web):** A front-end interface allows users to view recommended schedules, give feedback, and track performance. The interface connects to the scheduling engine through APIs or wireless modules for seamless interaction.

2. **Software Components:**

- **Chronotype Analyzer:** A preprocessing component that generates heatmaps based on collected user data, serving as input priors for the contextual bandit.

- **Feedback Processor:** Responsible for collecting and interpreting user responses, converting them into reward signals for the bandit model.

- **Backend Implementation**: Control logic of application and communicate with database for store application data and other cognitive load related data

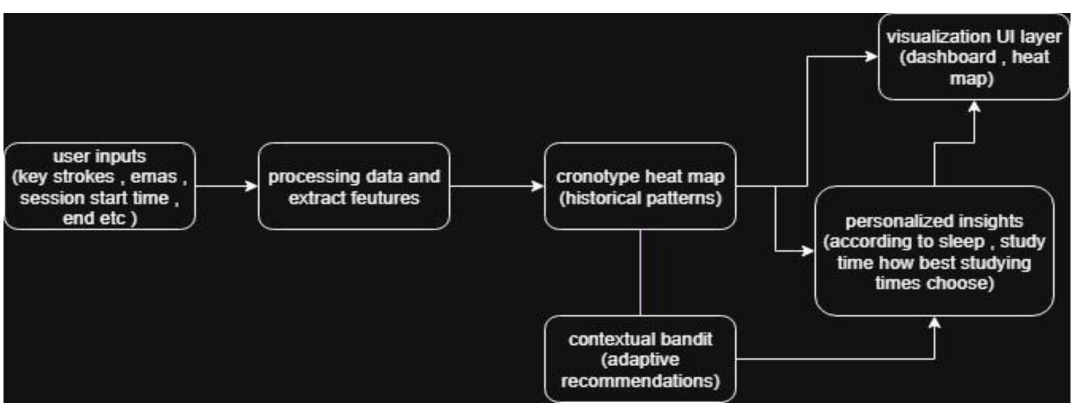

Figure 3. 2 – system diagram of cronotype predicter component

3.2 Implementation Steps

1. **Data Collection & Preprocessing:**

- **Collect User Activity:** Typing/scrolling patterns, session start/end times, micro-EMAs ("How focused are you?" ratings). Optional context like sleep duration, breaks, and session length.

- **Aggregate Historical Data:** To initiate study session optimized timers , developers need to collect cronotype data from present student using surveys and cronotype related web sites

2. **Chronotype Heatmap Module:**

- **Implement Algorithms :** Implment algorithms to visualize user productivity patterns

- **Insight generatation:** Generate insights according to system outputs

3. **Contextual Bandit Model:**

- **Choose Contextual Bandit Algorithm :** Choose a contextual bandit algorithm (e.g., LinUCB, Thompson Sampling).
- **Initialize Bandits Priors :** Initialize bandit priors using chronotype heatmap data to reduce unnecessary exploration.

4. **User Interface Implementation:**

- **Design the dashboard to display :** Heatmaps, personalized schedules, and adaptive insights.
- **Allow users to manually adjust preferences and view historical trends.**

5. **Optimization and Refinement:**

- **Performance Tuning:** Analyze field test results to identify any performance issues or limitations. Optimize the firmware and hardware settings to enhance communication reliability and coverage.

- **User Feedback:** Incorporate feedback from safari guides and tourists to refine the user interface and overall system functionality. Adjust based on real-world usage scenarios.

6. **Testing and Evaluation:**

- **Refine Algorithms :** Tune algorithms according to user inputs and cognitive load data to give more accurate insights and visualization.

- **Testing :** Testing system with system, unit, integration testing vice and test system usage with real users to evaluate performance

4. Project Requirements

4.1 Functional Requirements

Hardware Components:

- **Mobile/PC Devices:**
  - Smartphones (Android/iOS) or PCs: Devices used by users to interact with the study timer and provide activity data.

- **Optional Sensors (for Mobile):**
  - Accelerometer / Gyroscope: Detect device movement patterns.
  - Touchscreen / Keyboard Input: Capture typing speed, errors, and scrolling behavior

Software Components:

- **Vs Code / Android Studio:**
  - **Programming Environment:** Used for writing software implementation

- **Study Timer App:**
  - Application for mobile or desktop that monitors user activity, schedules study sessions, and collects micro-EMA responses.

- **Backend / Local Storage:**
  - Database or local storage for storing session data, focus scores, heatmap data, and contextual information (sleep, session length).

- **Machine Learning Module:**
  - Contextual bandit algorithm to adaptively recommend study sessions and breaks based on user state and historical behavior.
  - Chronotype analysis module to generate heatmaps and detect peak focus times.

4.2 Non-Functional Requirements

- **Privacy & Security:**
  - All user data should remain on-device unless the user explicitly opts in for cloud sync.

- **Adaptivity:**
  - Real-time adaptation to user cognitive state using contextual bandit recommendations

- **Usability:**
  - User-friendly interface for session scheduling, heatmap visualization, and EMA prompts.
  - Minimal interruption during study sessions

- **Reliability:**
  - Accurate capture of key activity metrics and timely micro-EMA prompts.

- **Performance & Efficiency:**
  - Lightweight data processing for smooth performance on mobile and desktop.
  - Minimal battery and resource usage for mobile devicess

4.3 Expected Test Cases

Data Capture Testing:

- **Objective:** Verify accurate recording of typing speed, scrolling behavior, session times, and EMA responses.
- **Procedure:** Simulate study sessions and check that all activity metrics are logged correctly.

Chronotype Heatmap Testing:

- **Objective:** Ensure heatmaps accurately reflect user productivity patterns over hours/days.
- **Procedure:** Compare generated heatmaps against known session performance trends.

Contextual Bandit Recommendation Testing:

- **Objective:** Validate that adaptive scheduling responds correctly to user state and past behavior.
- **Procedure:** Simulate different focus and fatigue scenarios and evaluate recommended study/break times.

User Interface Testing:

- **Objective:** Assess usability, clarity, and responsiveness of the app/dashboard.

- **Procedure:** Conduct user testing and collect feedback for interface improvements.

Integration Testing:

- **Objective:** Verify that all modules (data collection, heatmap, bandit, UI) work seamlessly together.
- **Procedure:** Run end-to-end sessions and confirm proper functioning of activity monitoring, heatmap generation, and adaptive recommendations.

5.Budget and Budget Justification

The budget for this project includes the costs for hardware components, software licenses, and personnel involved in the development and testing of the system. Below are the estimated costs to develop a prototype for the implementation.

	Description	Cost
Hardware Components	Mobile Deveices and sensors	Rs 0
Software Components	Vs code	Open-source
	Android Studio	Open-source
	Libraries and SDKs	Open-source

Table 5. 1 - Estimated Budget Allocation

6.Work Breakdown Structure

Phase & Task	Duration (weeks)	Deliverables
1. Planning & Literature Review	2	Annotated bibliography on chronotype studies, cognitive load estimation, contextual bandits; refined problem statement
2Sensor & Logging Design	3	Finalized logging specification (keystrokes, scrolls, session start/end times, micro-EMA prompts); privacy and consent protocol
3. Heatmap & Scheduler Prototype Development	5	Initial app capturing activity metadata; micro-EMA UI; local storage; prototype chronotype heatmap; contextual bandit skeleton
4. Pilot Data Collection	4	Pilot study with 3–5 users; preliminary dataset; feedback on usability and timing of prompts
5. Feature Engineering & Model Prototyping	6	Scripts for extracting features (typing speed, session duration, focus scores); baseline contextual bandit models; evaluation metrics report
6. Full-Scale Study Implementation	6	Recruitment of ~30 participants; deployment of app; ongoing data collection and monitoring
7. Data Analysis & Model Refinement	6	Personalized models trained; statistical analysis of correlation with micro-EMA responses; heatmap refinement; bandit performance report
8. User Evaluation & Interviews	4	Survey and interview instruments; coded themes; user burden assessment
9. Integration & API Development	4	API delivering real-time cognitive load scores and heatmap updates; integration with study timer app; documentation
10. Final Report & Dissemination	4	Research paper/report; poster; presentation of findings; recommendations for further development

Table 6.1 Work breakdown Structure for the Project Timeline

7. Ability of Commercialization / Entrepreneurship Potential

The proposed personalized study timer and chronotype heatmap system exhibits significant potential for commercialization due to its unique integration of behavioral insights, adaptive recommendations, and real-time productivity tracking. By leveraging contextual bandit models and chronotype-based analysis, the system delivers personalized study schedules and actionable insights that are not available in conventional study tools.

Key factors demonstrating its business potential include:

- Market Demand**:** Increasing numbers of students and professionals seek tools to optimize productivity and focus, particularly in remote learning and hybrid work environments.

- Unique Value Proposition: Unlike standard study applications, this system combines machine learning-driven adaptive scheduling, micro-EMA feedback, and chronotype analysis to provide tailored study recommendations.

- Scalability**:** The platform can be deployed across multiple platforms (desktop, mobile, web) and integrated with existing educational tools, expanding reach and adoption.

- Monetization Opportunities**:** Potential revenue streams include subscription models, premium analytics features, partnerships with educational institutions, and integration with learning management systems.

- User Benefits: Users gain actionable insights such as optimal study times, the effect of sleep on focus, and session-length recommendations, leading to improved productivity and enhanced learning outcomes.

Overall, the system addresses a critical gap in personalized study tools, offering both tangible benefits to users and a scalable product with strong market viability.

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